SAMPLE

Neutrinos

The Ghostly Particles

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Introduction

A neutrino (new-TREE-noh; denoted by the Greek letter ν) is an elementary particle that interacts via the weak interaction and gravity. The neutrino is so named because it is electrically neutral and because its rest mass is so small (-ino) that it was long thought to be zero. The rest mass of the neutrino is much smaller than that of the other known elementary particles (excluding massless particles).

The weak force has a very short range, the gravitational interaction is extremely weak due to the very small mass of the neutrino, and neutrinos do not participate in the electromagnetic interaction or the strong interaction. Consequently, neutrinos typically pass through normal matter unimpeded and with no detectable effect. Weak interactions create neutrinos in one of three leptonic flavors: electron neutrino, νe; muon neutrino, νμ and tau neutrino, ντ. Each flavor is associated with the correspondingly named charged lepton.

Although neutrinos were long believed to be massless, it is now known that there are three discrete neutrino masses with different values (all tiny, the smallest of which could be zero), but the three masses do not uniquely correspond to the three flavors: A neutrino created with a specific flavor is a specific mixture of all three mass states (a quantum superposition). Similar to some other neutral particles, neutrinos oscillate between different flavors in flight as a consequence.

For example, an electron neutrino produced in a beta decay reaction may interact in a distant detector as a muon or tau neutrino. The three mass values are not yet known as of 2026, but laboratory experiments and cosmological observations have determined the differences of their squares, an upper limit on their sum (< 0.120 eV/c2), and an upper limit on the mass of the electron neutrino. Neutrinos are fermions, which have spin of 1/2ħ.

The exceptionally feeble interaction with normal matter allowed the neutrinos to pass through the churning mass of the exploding star, while the electromagnetic photons were slowed.

A neutrino detector is a physics apparatus which is designed to study neutrinos. Because neutrinos only weakly interact with other particles of matter, neutrino detectors must be very large to detect a significant number of neutrinos.

Chapter 1: The Discovery of Neutrinos

These include the beta decay of atomic nuclei or hadrons and natural nuclear reactions such as those that take place in the core of a star. Other mechanisms include artificial nuclear reactions in nuclear reactors, nuclear bombs, or particle accelerators, supernovas, during the spin-down of a neutron star and when cosmic rays or accelerated particle beams strike atoms. The majority of neutrinos which are detected about the Earth are from nuclear reactions inside the Sun.

At the surface of the Earth, the flux is about 65 billion (6.5×1010) solar neutrinos, per second per square centimeter. Neutrinos can be used for tomography of the interior of the Earth. The neutrino was postulated first by Wolfgang Pauli in 1930 to explain how beta decay could conserve energy, momentum, and angular momentum (spin).

In contrast to Niels Bohr, who proposed a statistical version of the conservation laws to explain the observed continuous energy spectra of electrons in beta decay, Pauli hypothesized an undetected particle that he called a "neutron", using the same -on ending employed for naming both the proton and the electron. He considered that the new particle was emitted from the nucleus together with the beta particles (electrons or positrons) and had a mass similar to the electron.

James Chadwick discovered a much more massive neutral nuclear particle in 1932 and named it a neutron also, leaving two kinds of particles with the same name. The word "neutrino" entered the scientific vocabulary through Enrico Fermi, who used it during a conference in Paris in July 1932 and at the Solvay Conference in October 1933, where Pauli also employed it.

The name (the Italian equivalent of "little neutral one") was jokingly coined by Edoardo Amaldi during a conversation with Fermi at the Institute of Physics of via Panisperna in Rome, in order to distinguish this light neutral particle from Chadwick's heavy neutron.

In Fermi's theory of beta decay, Chadwick's large neutral particle could decay to a proton, electron, and the smaller neutral particle (now called an electron antineutrino): Fermi's paper, written in 1934, unified Pauli's neutrino with Paul Dirac's positron and Werner Heisenberg's neutron-proton model and gave a solid theoretical basis for future experimental work.

By 1934, there was experimental evidence against Bohr's idea that energy conservation is invalid for beta decay: At the Solvay conference of that year, measurements of the energy spectra of beta particles (electrons) were reported, showing that there is a strict limit on the energy of electrons from each type of beta decay. Such a limit is not expected if the conservation of energy is invalid, in which case any amount of energy would be statistically available in at least a few decays.

The natural explanation of the beta decay spectrum as first measured in 1934 was that only a limited (and conserved) amount of energy was available, and a new particle was sometimes taking a varying fraction of this limited energy, leaving the rest for the beta particle. Pauli made use of the occasion to publicly emphasize that the still-undetected "neutrino" must be an actual particle.

The first evidence of the reality of neutrinos came in 1938 via simultaneous cloud-chamber measurements of the electron and the recoil of the nucleus. In 1942, Wang Ganchang first proposed the use of beta capture to experimentally detect neutrinos. In the 20 July 1956 issue of Science, Clyde Cowan, Frederick Reines, Francis B. Kruse, and Austin D. McGuire published confirmation that they had detected the neutrino, a result that was rewarded almost forty years later with the 1995 Nobel Prize.

In this experiment, now known as the Cowan-Reines neutrino experiment, antineutrinos created in a nuclear reactor by beta decay reacted with protons to produce neutrons and positrons: The positron quickly finds an electron, and they annihilate each other. The two resulting gamma rays (γ) are detectable. The neutron can be detected by its capture on an appropriate nucleus, releasing a gamma ray. The coincidence of both events--positron annihilation and neutron capture--gives a unique signature of an antineutrino interaction.

In February 1965, the first neutrino found in nature was identified by a group including Frederick Reines and Friedel Sellschop. The experiment was performed in a specially prepared chamber at a depth of 3 km in the East Rand ("ERPM") gold mine near Boksburg, South Africa. A plaque in the main building commemorates the discovery. The experiments also implemented a primitive neutrino astronomy and looked at issues of neutrino physics and weak interactions.

The antineutrino discovered by Clyde Cowan and Frederick Reines was the antiparticle of the electron neutrino. In 1962, Leon M. Lederman, Melvin Schwartz, and Jack Steinberger showed that more than one type of neutrino exists by first detecting interactions of the muon neutrino (already hypothesised with the name neutretto), which earned them the 1988 Nobel Prize in Physics.

Neutrino detectors are often built underground, to isolate the detector from cosmic rays and other background radiation. The field of neutrino astronomy is still very much in its infancy - the only confirmed extraterrestrial sources as of 2018 are the Sun and the supernova 1987A in the nearby Large Magellanic Cloud. Another likely source (three standard deviations) is the blazar TXS 0506+056 about 3.7 billion light years away. Neutrino observatories will "give astronomers fresh eyes with which to study the universe".

Various detection methods have been used. Super Kamiokande is a large volume of water surrounded by phototubes that watch for the Cherenkov radiation emitted when an incoming neutrino creates an electron or muon in the water. The Sudbury Neutrino Observatory was similar, but used heavy water as the detecting medium. Other detectors have consisted of large volumes of chlorine or gallium which are periodically checked for excesses of argon or germanium, respectively, which are created by neutrinos interacting with the original substance.

Tau leptons decay essentially immediately to either another charged lepton or pions, and cannot be observed directly in this kind of detector.

At low energies, a neutrino can scatter from the entire nucleus of an atom, rather than the individual nucleons, in a process known as coherent neutral current neutrino-nucleus elastic scattering or coherent neutrino scattering. This effect has been used to make an extremely small neutrino detector.